# GeSANet: Geospatial-Awareness Network for VHR Remote Sensing Image Change Detection

Xiaoyang Zhao, Keyun Zhao, Siyao Li, and Xianghai Wang

***Abstract*— The characteristics of very high-resolution (VHR) remote sensing images (RSIs) have higher spatial resolution inherently and are easier to obtain globally compared with hyperspectral images (HSIs), making it possible to detect small-scale land cover changes in multiple applications. RSI change detection (RSI-CD) based on deep learning has been paid attention to become a frontier research field in recent years and is currently facing two challenging problems. The first is high dependence on registration between bitemporal images caused by high spatial resolution. The other is high pseudo-change information response caused by low spectral resolution. In order to address the abovementioned two problems, a novel RSI-CD framework called geospatial-awareness network (GeSANet) based on the geospatial position matching mechanism (PMM) with multilevel adjustment and the geospatial content reasoning mechanism (CRM) with diverse pseudo-change information filtering is proposed. First, the PMM assigns independent 2-D offset coordinates to each position in the previous temporal image. Afterward, bilinear interpolation is employed to obtain the subpixel feature value after the offset, and the sparse results based on the difference are transmitted to the next level prediction to realize multilevel geospatial correction. The CRM extracts the global features from the corrected sparse feature map in terms of dimensions, implementing effective discriminant feature extraction on the basis of the original feature map in a stepwise refinement manner through the cross-dimension exchange mechanism, to filter out various pseudo-change information as well as maintain real change information. Comparison experiments with five recent state-of-the-art (SOTA) methods are carried out on two popular datasets with diverse changes, and the results show that the proposed method has good robustness and validity for multitemporal RSI-CD. In particular, it has a strong comparative advantage in detecting small entity changes and edge details. The source code of the proposed framework can be downloaded from https://github.com/zxylnnu/GeSANet.**

***Index Terms*— Attention mechanism, change detection, deep learning, image registration, pseudo-change, very high-resolution (VHR) remote sensing image (RSI).**

## I. Introduction

REMOTE sensing image change detection (RSI-CD) aims at obtaining change information of entities or phenomena on land cover and land use from multitemporal RSIs at the same geographical location. In recent years, it has been valued in practical application fields of Earth observation, such as urban planning, environmental monitoring, agricultural investigation, disaster assessment, and map revision [1]. Very high-resolution (VHR) remote sensing, as a kind of passive optical remote sensing technology to get Earth observation information, captures three electromagnetic waves radiated by ground objects and obtains different band information of the same target at the same time in the form of photography or scanning. It can discriminate ground objects according to the differences in image morphology and structure, which expands the amount of telemetry information [2]. Compared with hyperspectral images (HSIs), VHR RSIs are easier to obtain in practical applications. Especially for some special terrain scenes, VHR RSIs generally have a higher spatial resolution to capture small objects. These advantages and characteristics of VHR RSIs make it possible for multiple applications of Earth observation and small-scale ground object change detection. In recent years, the fine change detection technology for VHR RSIs has become a frontier field of RSI information processing [3].

Early change detection methods gain the final change map mainly through extracting, analyzing, and comparing spatial or spectral information of the same position pixels between bitemporal images, roughly including algebra-, transformation-, and machine learning (ML)-based CD methods. Algebra-based CD methods are also known as direct comparison methods, which achieve change maps by setting the threshold according to the calculation formula. Common ones include image difference (ID) methods [4], image ratio (IR) methods [5], and change vector analysis (CVA) methods [6]. Compared with the previous two types of methods, CVA-type methods can make multivariate change detection (MAD) come true. Transformation-based CD methods are a kind of methods that measure the change information after transforming the image from spatial domain to a specific feature space, and such methods mainly include principal component analysis (PCA) methods [7], MAD methods [8], slow feature analysis (SFA) methods [9], and so on. PCA-type CD methods achieve the change map by enhancing change information in the image and eliminating interference information based on the PCA algorithm, and the MAD extracts change information by maximizing the variance of the difference map, while SFA improves the separability of changed and unchanged pixels by assuming that unchanged pixels have less change difference between bitemporal images. Although transformation-based CD methods can generally

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improve the detection accuracy to a certain extent compared with algebra-based CD methods, it is easy to mix some salt and pepper noise in the final detection result map. As ML methods are widely used in the RSI classification, ML-based CD methods have developed rapidly as well, such as support vector machine (SVM) methods [10], conditional random field methods [11], and artificial neural network methods [12], [13], which have emerged. An adaptive ANN [12] that provides a variable threshold for each pixel in difference map is proposed to improve the robustness and accuracy of the detection model. In fact, the development of ANN-based change detection techniques has laid the foundation for the success of later deep learning-based change detection methods.

The success of deep learning in the computer vision research field has also brought new ideas to RSI processing, achieving good results in the fields of classification [14], target detection [15], and so on. In recent years, the deep learning mechanism has been applied to the field of change detection and attracted much attention. There are generally two ways to realize the CD process up to now. One detection mode is networks based on "two steps" [16], that is, the initial detection result of the bitemporal images is obtained by making differences or other methods as the input and the final change information is subsequently acquired through the layerwise abstraction and refinement of the network. The advantages of this kind of method are that the initial detection in the early stage is easy to implement, and the network structure is relatively simple and easy to train. However, these nonend-to-end networks usually lose some available features, and the ability to get context information from images is very limited. The other detection mode is with dual branches as the main framework, called the Siamese network [17]. It takes bitemporal images as input, and multilevel feature maps are extracted by the backbone network to produce a more discriminative feature map through mechanisms such as attention. Although such global inference methods are capable of adaptively obtaining change information in an end-to-end way, their excessive introduction of background information increases the instability and computational cost.

Although a large number of CD methods have been proposed until now, the development of this research field still faces the following challenging problems. First, CD, as a time sequence problem, involves registration problems between bitemporal images inevitably. Despite multitemporal HSI-CD also involving this problem, due to the relatively high spatial resolution of VHR RSIs, the problem is more prominent in VHR RSI-CD. At present, the registration error between bitemporal VHR RSIs is widespread and has not been solved for good. There is still a lack of explicit modeling of the registration problem in the CD scheme. Second, the problem of the excessive response to pseudo-change (see Section II-A for details) is a common occurrence during a CD process due to the characteristics of VHR RSIs. It affects the accuracy of CD results greatly. How to effectively suppress the pseudo-change information has always been a difficult problem in the VHR RSI-CD research field. Based on these aforementioned problems, the geospatial-awareness network (GeSANet) for VHR RSI-CD is proposed, from the perspective of suppressing pseudo-change information. The proposed GeSANet can reduce the interference due to misregistration and pseudo-change information caused by diverse manual and natural factors. The main contributions of this article are reflected in the following aspects.

1) A geospatial position matching mechanism (PMM) based on multilevel correction is proposed. Each pixel feature in the previous temporal image is assigned independent 2-D offset coordinates, and the subpixel values are obtained through bilinear interpolation. Also, the sparse result based on the difference is transferred to the next level to achieve multilevel geospatial position correction. This mechanism breaks through the traditional inherent one-to-one mapping pattern of spatial position and realizes the pixel-level adaptive matching of multitemporal images based on geospatial position.
2) A geospatial content reasoning mechanism (CRM) based on filtering out diverse pseudo-change information is proposed. According to the tensor CANDECAMP/PARAFAC (CP) decomposition theory, global features are extracted from the corrected sparse feature map in dimensions and used as the initial vectors to be reconstructed. Furthermore, the effective discriminative feature extraction based on the original feature map is realized in a gradual refinement manner through the cross-dimensional exchange mechanism, to achieve the purpose of filtering out multiple pseudo-change information and retaining real change information at the same time.
3) Based on the abovementioned PMM and CRM, the GeSANet for VHR RSI-CD is proposed. Compared with multiple state-of-the-art (SOTA) methods proposed in the past three years, the GeSANet achieves higher effectiveness with more accurate detected edges and higher sensitivity to small-scale entity changes.

The rest of this article is organized as follows. Section II analyzes the related work, including the characteristic analysis of VHR RSIs and the analysis of bitemporal VHR RSI-CD with the Siamese framework. Section III describes the CD framework proposed in this article, including the overall structure of the GeSANet, PMM, CRM, and loss function. Section IV presents the experimental results and the corresponding ablation experiment analysis, mainly including the introduction to datasets and evaluation metrics, comparison algorithms and experimental environment, LEVIR dataset and CDD dataset experimental results and analysis, as well as ablation experiments and discussion. Section V provides the conclusion.

## II. Related Work

### A. Characteristics of VHR RSIs and the Influence on CD Task

The mechanism of RS sensor imaging equipment determines that the current RS hardware technical conditions are still unable to acquire multiband RSIs with both high spatial resolution and high spectral resolution at the same time [18].

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Under the same conditions such as incident light energy and signal-to-noise ratio, generally, the higher the spectral resolution of the sensor, the lower its spatial resolution, and vice versa. VHR RSIs are a special case of MSIs. VHR RSIs have only three spectral bands but higher spatial resolution. The spatial resolution of some popular public VHR RSI datasets can reach meter level or even submeter level [19], [20], [21].

The high spatial resolution of VHR RSI makes it contain rich geometric structure and fine texture information of ground objects, thus providing strong support for detecting changes in small-scale targets. However, the above characteristics of VHR RSIs also bring a certain negative impact on CD, which is specifically manifested in the following aspects.

*1) High Responsivity to Pseudo-Changes:* The low spectral resolution of VHR RSI makes it impossible to provide sufficient spectral discriminative information under the imaging conditions of season alternation, illumination change, shadow change, and various interference factors, resulting in the detection results being highly prone to discontinuity at positions such as the boundary of ground objects, so that the detection results are mixed with some pseudo-change information.

*2) Severe Reliance on High-Precision Registration Algorithms:* Since change detection belongs to the multitemporal analysis task, it is usually necessary to ensure spatial position registration with a difference of less than 0.5 resolution between images [22]. In practical applications, the registration between multitemporal RSIs is very difficult to meet the requirements of the standard registration error, and this factor has also become an important reason for limiting change detection accuracy.

*3) Spatial Information Redundancy:* The fine ground sampling capability of VHR RSI not only obtains clear visual image representation but also makes the image have strong local similarity and certain nonlocal similarity, thus forming a large amount of redundancy of spatial information.

### B. Attention Mechanisms Used in VHR RSI-CD Networks

The powerful fitting capability of deep learning can adaptively learn the loss between input and output. Considering that bitemporal CD involves two inputs, the Siamese network structure is naturally considered. It uses a pretrained backbone to extract multiscale deep features and then obtains the change result map through supervised training to realize an end-to-end automatic detection.

Most of the CD methods based on deep learning in recent years focus on the reasonable introduction of attention mechanisms [23] under the Siamese network structure to obtain a denser affinity matrix to encode the global correlation of images, thereby enhancing the capability of the network to discriminate change information. It mainly includes the following aspects of exploration.

*1) Effect of Channel Dimension Features on Detection Results:* Fang et al. [24] proposed a densely connected Siamese network with an ensemble channel attention module (ECAM), which improves the detection accuracy at edge position by bridging the semantic gap between deep and shallow features.

*2) Effect of Spatial–Channel Joint Dimension Features on Detection Results:* Zhang et al. [17] and Chen et al. [25] proposed a dual attention mechanism, respectively, to obtain more discriminative feature representation and achieve finer detection.

*3) Effect of Temporal Dimension Features on Detection Results:* Chen et al. [26] proposed a dynamic receptive temporal attention network to analyze the effect of the size of temporal attention range size on the detection performance.

*4) Effect of Spatial–Temporal Joint Dimension Features on Detection Results:* Chen and Shi [19] proposed a spatial–temporal attention model that obtains the relationship between each pixel and other pixels in space by stacking bitemporal deep features.

It can be seen that most of the current change detection networks rely on various self-attention models to calculate the affinity relationship of channel, spatial, and temporal dimensions. However, redundant feature calculation undoubtedly brings high computational costs.

### C. Modeling the Relationship Between Bitemporal Images

Modeling the relationship between bitemporal images is one of the main characteristics to distinguish between CD and classification. The modeling methods can be divided into the following three categories. The first is using simple information fusion methods, such as concatenation [17], [27] and difference [28] operations. However, the measures taken by the methods are so simplistic that they do not have the ability to explore deep temporal correlations. The second is to use an attention-type mechanism introduced in Section II-B to explore the degree of correlation between pixels at any two positions in temporal images, generally including various variants centered on temporal attention [26], [29], including cross attention and spatial–temporal attention. However, these methods often require a heavy computational cost. The third type of method is to use networks that process sequential data, such as recurrent neural network (RNN)/long short-term memory (LSTM). Some research works attempt to combine CNN and RNN organically [30] and integrate them into a unified framework, where the former is used to extract deep abstract features in images and the latter is used to model temporal correlation. Also, these methods always use small image patches as input (such as $5 \times 5$), which is not large enough to represent a complete ground object. Therefore, it limits the capability of networks to detect and perceive the knowledge that contains in images severely, and it is impossible to realize the extraction of mid- and long-distance information.

Furthermore, the scenarios applicable to the above three mentioned-type methods are usually based on high-precision registration between bitemporal images. When misregistration error occurs, the above networks are often not robust enough. Although the self-attention-based models can alleviate this problem to some extent, due to the huge amount of calculation of matrix multiplication, it limits the learning content of each batch of the network. Therefore, image registration, which can reduce the shift of geospatial positions, is one of the most important preprocessings for handling multitemporal-oriented

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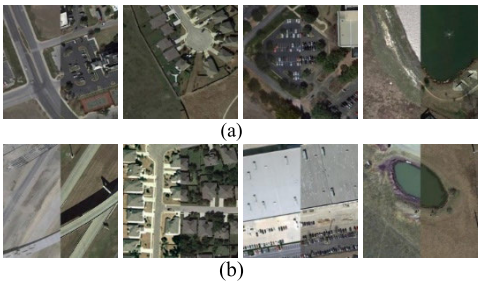


Fig. 1. Accurate registration and misregistration samples of the LEVIR dataset. (a) Accurate registration samples. (b) Misregistration error samples.

image change detection and has always been a research hotspot and difficult problem. In the field of change detection, the accuracy of registration directly affects the detection result. To this end, some datasets have been specifically published to overcome the misregistration problem that arises in change detection. Fig. 1 shows several registration samples in the LEVIR dataset provided by Chen and Shi [19]. The accurate registration samples are given in the first row, and the second row demonstrates misregistration error samples. It is clear that misregistration error is widespread and more evident at the edges of ground objects, which can easily bring negative effects and difficulties to network learning.

Some studies [31], [32], [33] have achieved the goal of converting the receptive field by introducing the deformable convolution proposed by Dai et al. [34] and Zhu et al. [35]. However, these methods take advantage of the ability of deformable convolution to warp the extracted features, but they adopt the pattern of fusing bitemporal images into a whole, that is, warping the features of bitemporal images using the same offset, completely ignore the exploitation of the positional correspondence between bitemporal images, and cannot achieve the purpose of feature-level registration. It can be seen that how to apply the deformable convolution idea to model the bitemporal relationship in the field of change detection remains to be further studied.

## III. Methodology

In this section, the overall structure of the proposed network is expounded, and the PMM and CRM are explained specifically in turn.

### A. Overall Structure of GeSANet

Based on the previous analysis, a novel GeSANet for VHR RSI-CD oriented framework is proposed, which enables the network to learn a mapping relationship from dual-input RSIs to ground truth (GT) through end-to-end fully supervised training and assigns unique changed or unchanged prediction label to each spatial element. The overall structure of GeSANet is shown in Fig. 2.

Given that bitemporal VHR RSIs $X_{T1} \in R^{H\times W\times B}$ and $X_{T2} \in R^{H\times W\times B}$ are set to be the input of the network, where $H$ and $W$ are the height and width of each band, respectively, $B$ represents the number of bands, and the corresponding GT is $Y \in R^{H\times W\times B}$, logically speaking, the present network consists of the following three main components: feature extraction module, PMM, and CRM.

1) Under the Siamese framework, multiscale feature groups $F_{T1}^{l} = \{F_{T1}^{1}, \ldots, F_{T1}^{4}\}$ and $F_{T2}^{l} = \{F_{T2}^{1}, \ldots, F_{T2}^{4}\}$ of $X_{T1}$ and $X_{T2}$ are extracted using the pretrained Resnet18 network (refer to Fig. 2) as the encoders of the network, where $l = 1, 2, 3, 4$ represents the level number of multiscale.
2) Take the above $F_{T1}^{l}$ and $F_{T2}^{l}$ as the input of PMM, for obtaining multiscale corrected feature maps $F_{\text{PMMs}}^{l}$ and densely connected feature $F_{\text{CP}}$. $F_{\text{CP}}$ is a combination of a set of feature maps, and each feature map is obtained by a convolution operation with the stride of two and followed by a concatenation with the deeper feature map. Then, decode $F_{\text{CP}}$ by using the UpSample Layer to gain a dense fusion feature $F_{\text{PMM}}$, which is going to be the residual content of CRM.
3) Use $F_{\text{PMM}}$ as the input of CRM as well and reasoning effective change feature $F_{\text{CRM}}$ from it.
4) The fusion feature $F_{\text{PMM}}$ and effective change feature $F_{\text{CRM}}$ are fed into the rescale layer and classifier layer simultaneously, to restore them to the original input image size. Finally, the summed result is regarded as the output of the network.

### B. Position Matching Mechanism

To match the offset $F_{T1}^{l}$ feature map with $F_{T2}^{l}$ in spatial position, the PMM shown in Fig. 3 assigns independent offset coordinates to each element in a $3 \times 3$ kernel.

For multiscale feature maps $F_{T1}^{l}$ and $F_{T2}^{l}$ of bitemporal VHR RSIs, the process of obtaining multiscale corrected residual feature map $F_{\text{PMMs}}^{l}$ can be described as follows:

$$F_{\text{PMMs}}^{l} = F_{T2}^{l} - F_{\Delta p}^{l} = F_{T2}^{l} - \text{Bi}\left(\text{Conv2d}\left(F_{T1}^{l}\right)\right) \tag{1}$$

where $\text{Bi}(\cdot)$ is a bilinear interpolation kernel function. It can be seen from (1) that PMM can be divided into two subproblems: estimating offset coordinates and calculating offset values.

*1) Estimating Offset Coordinates:* For the current feature map $F_{T1}^{l}$, centered at $p_0$, with the convolutional kernel size of ks $\times$ ks, PMM provides a set of offset $\Delta p$, if $K = \{(-((\text{ks}-1)/2), -((\text{ks}-1)/2)), \ldots, (((\text{ks}-1)/2), ((\text{ks}-1)/2))\}$ is used to represent coordinates of plain convolutional sampling grid, the corrected feature map of $T1$ temporal feature map can be expressed as

$$F_{\Delta p}^{l}(p_0) = \sum_{p_n \in K} w(p_n)\left[F_{T1}^{l}(p_0 + p_n + \Delta p_n) \cdot \text{sig}(\Delta m_n)\right] \tag{2}$$

where $w$ describes weighting coefficients, $p_n$ enumerates all position coordinates in $K$, and $\Delta m_n$ is employed for controlling the degree of offset since each element in the kernel has different contributions to the output features. An additional $\text{sig}(\cdot)$ is required, which stands for the Sigmoid function, mapping $\Delta m_n$ into (0, 1), to make the kernel focus on as

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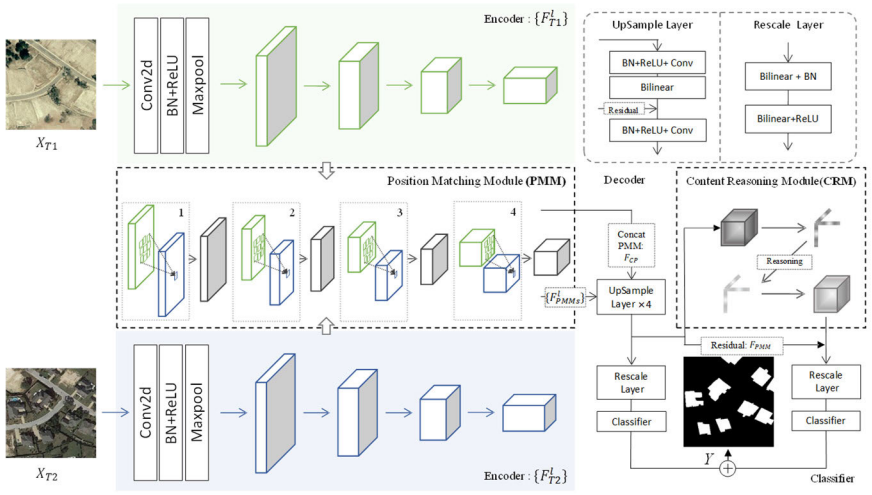



Fig. 2. Structure diagram of GeSANet.

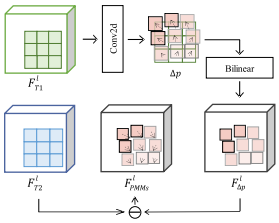



Fig. 3. Schematic of PMM.

many effective or significant feature points as possible

$$\begin{cases} \Delta p = \{\Delta p_n\} = \left\{(\Delta x_n, \Delta y_n)_{n=1}^{N} \mid N = \text{ks} \times \text{ks}\right\} \\ \quad \Delta p \in R^{2N\times 1\times 1} \\ \Delta m = \{\Delta m_n \mid n = 1, 2, \ldots, N\} \\ \quad \Delta m \in R^{N\times 1\times 1}. \end{cases} \tag{3}$$

*2) Calculating Offset Values:* On account of $\Delta p$ learned by the network that is usually discrete and fractional, we thus adopt the same bilinear interpolation method with backpropagating gradient as in [34] to calculate the offset values.

$$\begin{cases} F_{T1}^l(p) = \sum_q \text{Bi}(q, p) \cdot F_{T1}^l(q) \\ p = p_0 + p_n + \Delta p_n \end{cases} \tag{4}$$

where $p$ signifies the position of any subpixel in $F_{T1}^l$, $q$ enumerates the integer sampling grid on the corresponding spatial position, and $\text{Bi}(\cdot)$ is the bilinear interpolation kernel function, i.e.,

$$\begin{cases} \text{Bi}(q, p) = g(q_x, p_x) \cdot g\left(q_y, p_y\right) \\ g(a, b) = \max(0, 1 - |a - b|). \end{cases} \tag{5}$$

### C. Content Reasoning Mechanism

By introducing the knowledge of CP decomposition and reconstruction in tensor complementation [36], [37], it is well known that for any third-order tensor $\boldsymbol{y} \in R^{M\times N\times T}$, if $r(\boldsymbol{y}) = R$, $\forall y_{ijt} \in \boldsymbol{y}$ can be expressed as

$$\begin{cases} y_{ijt} \approx \sum_{r=1}^{R} u_{ir} \cdot v_{jr} \cdot x_{tr} \\ i = 1, \ldots, M; \quad j = 1, \ldots, N; \ t = 1, \ldots, T \end{cases} \tag{6}$$

where $\boldsymbol{U} \in R^{M\times R}$, $\boldsymbol{V} \in R^{N\times R}$, and $\boldsymbol{X} \in R^{T\times R}$ are the factor matrices of CP decomposition, any element in the factor matrix is denoted as $u_{ir}$, $v_{jr}$, and $x_{tr}$, and any column vector is denoted as $\boldsymbol{u}_r \in R^M$, $\boldsymbol{v}_r \in R^N$, and $\boldsymbol{x}_r \in R^T$. Therefore, CP decomposition can also be written as

$$\boldsymbol{y} \approx [\boldsymbol{U}, \boldsymbol{V}, \boldsymbol{X}] \equiv \sum_{r=1}^{R} \boldsymbol{u}_r \circ \boldsymbol{v}_r \circ \boldsymbol{x}_r. \tag{7}$$

Here, $\circ$ denotes the outer product of vectors.

The physical meaning of (7) is that any high-rank tensor can be represented by a series combination of single-rank vectors. To this end, CRM represented in Fig. 4 is constructed, and the effective change feature $F_{\text{CRM}}$ is deduced based on the input-dense fusion feature map $F_{\text{PMM}}$.

The proposed CRM is composed of three parts: multidimensional single-rank vector acquirement, nonlinear mapping, and high-rank tensor reconstruction. The specific operations are given as follows.

1) *Multidimensional Single-Rank Vector Acquirement:* For obtaining comprehensive features in each dimension of $F_{\text{PMM}}$, the global average pooling (GAP) operation is employed. Single-rank vector features of channel dimension $F_C$, height dimension $F_H$, and width dimension $F_w$ are set as the initial content to be inferred.
2) *Nonlinear Mapping:* After obtaining multidimensional single-rank vectors, the desired real change vector features are inferred through nonlinear mapping layers with convolution operations followed by the Sigmoid function, as $\boldsymbol{u}_r$, $\boldsymbol{v}_r$, and $\boldsymbol{x}_r$ in (7). The Sigmoid function used here is not only to enhance the nonlinear expression of this module but also to calculate the importance of each element in the vector, inspired by SENet [38], achieving the purpose of activation or inhibition.

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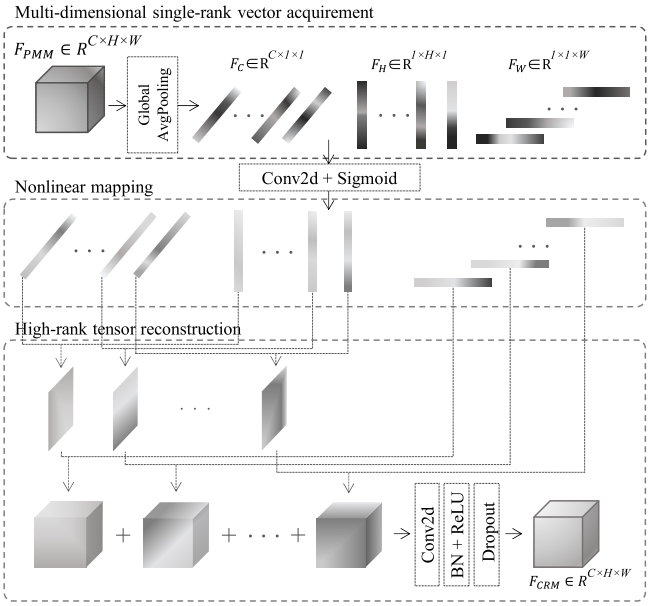



Fig. 4. Schematic of CRM.

3) *High-Rank Tensor Reconstruction:* Finally, the high-rank tensor containing clean change features is reconstructed by the cross-dimensional exchange product shown in (7).

### D. Loss Function

During the training process, given bitemporal VHR RSIs and the corresponding GT, denoted as $\{(X_{T1}^i, X_{T2}^i), Y^i\}_{i=1}^N \in R^{H\times W\times C}$, where $N$ is the total number of training samples, the change map is obtained by optimizing the loss function and adjusting the network parameters step by step. The binary cross-entropy (BCE) loss function defined in (8) is adopted for calculating the loss

$$L_{\text{BCE}} = -\frac{1}{H\times W}\sum_{p=1}^{H\times W}\left(Y_p \log\left(\hat{Y}_p\right) + \left(1-Y_p\right)\log\left(1-\hat{Y}_p\right)\right) \tag{8}$$

where $Y_p$ is the label of $p$th pixel of GT. If $Y_p = 1$, it signifies that the pixel belongs to the changed class. On the contrary, if $Y_p = 0$, the pixel belongs to the unchanged class. $\hat{Y}_p$ indicates the probability of $p$ belonging to the changed class on the predicted change map, while $1-\hat{Y}_p$ is the probability of $p$ belonging to the unchanged class.

Based on the discussion of each part, the implementation of GeSANet can be summarized in Algorithm 1.

**Algorithm 1** Flowchart of GeSANet

**Input:**

Bi-temporal VHR RSIs: $\left\{\left(X_{T1}^i, X_{T2}^i\right)\right\}_{i=1}^N$, GT: $\left\{Y^i\right\}_{i=1}^N$; learning rate $lr$, training epochs $E$, Adam optimizer;

**Output:**

Change map: $\left\{F_{CM}^i\right\}_{i=1}^N$;

**for** each *epoch* $\in [1, E]$ **do**

1: Load pre-trained Resnet18:

1.1: Extract multiscale feature group $\left\{F_{T1}^l\right\}^i$ from $X_{T1}^i$: $\left\{F_{T1}^l\right\}^i = \left\{\left\{F_{T1}^1, \ldots, F_{T1}^4\right\}\right\}^i$;

1.2: Extract multiscale feature group $\left\{F_{T2}^l\right\}^i$ from $X_{T2}^i$: $\left\{F_{T2}^l\right\}^i = \left\{\left\{F_{T2}^1, \ldots, F_{T2}^4\right\}\right\}^i$;

2: Correct offsets:

After PMM mentioned in (1)∼(5),

2.1: Obtain multi-scale corrected feature maps $\left\{F_{\text{PMMs}}^l\right\}^i$: $\left\{F_{\text{PMMs}}^l\right\}^i = \left\{\left\{F_{\text{PMMs}}^1, \ldots, F_{\text{PMMs}}^4\right\}\right\}^i$;

2.2: Gain densely connected feature $F_{CP}$;

3: After UpSample Layer shown in Fig. 2,

Gain dense fusion feature $F_{PMM}^i$;

4: Reason changes:

After CRM corresponding to (6)∼(7),

Gain effective change feature map $F_{CRM}^i$;

5: Gain final change map $F_{CM}^i$:

5.1: After $F_{PMM}^i$ and $F_{CRM}^i$ going through Rescale and Classifier Layer, Gain $F_{CM1}^i$ and $F_{CM2}^i$;

5.2: $F_{CM}^i = F_{CM1}^i + F_{CM2}^i$;

6: Calculate the loss $L_{BCE}^i$ between $F_{CM}^i$ and $Y^i$, using BCE given in (9);

7. Back-propagate $L_{BCE}^i$, and solve it with Adam;

**end for**

## IV. Experiments and Analysis

### A. Datasets and Evaluation Metrics

In order to verify the effectiveness of the proposed network, a series of comparison and ablation experiments is carried out on two public submeter-level RSI-CD datasets: LEVIR [19] and CDD [20]. The details of these two datasets and the five evaluation metrics used in this article are given as follows.

*1) Datasets:* The following holds.

*a) LEVIR:* This dataset belongs to a large-scale building binary CD dataset, which is composed of 637 pairs of bitemporal VHR RSIs with 1024 × 1024 size and 0.5-m/pixel spatial resolution. It covers multiscale building-related changes in more than 20 regions from 2002 to 2018. The detection difficulty is increased by introducing various change types, such as misregistration, season alternation, and illumination change.

*b) CDD:* As one of the most commonly used datasets in this research field, it mainly contains three types of images with spatial resolution ranging from 3 to 100 cm/pixel: composite images without relative displacement of objects, composite images with small relative displacements of objects, and real season changing. After cropping, there are 10 000 and 3000 pairs of bitemporal VHR RSIs with 256 × 256 size for training and validation, respectively, and another 3000 pairs for testing. The scale of this dataset is very large, with extremely obvious differences caused by season changing and various change types.

Fig. 5 shows some test examples of the datasets mentioned above. The LEVIR primarily encompasses the detection requirements for large-size, multiscale, all-around building changes, with the challenge of pseudo-changes suppression such as misregistration, season alternation, and shadow

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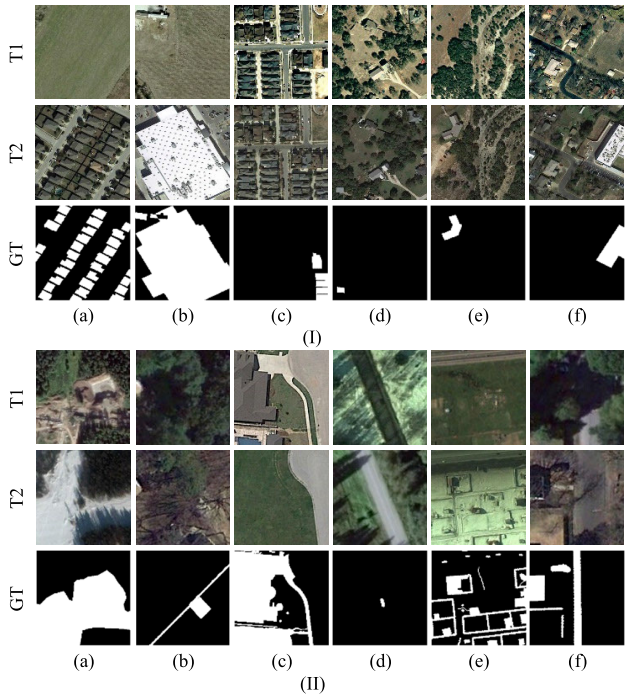



Fig. 5. Test examples of LEVIR and CDD datasets. (I) LEVIR. (II) CDD. (a) Sapmle 1. (b) Sapmle 2. (c) Sapmle 3. (d) Sapmle 4. (e) Sapmle 5. (f) Sapmle 6.

changing, while the changes in CDD are more diverse, including not only the increase or decrease of various buildings but also multiple LULC changes.

*2) Evaluation Metrics:* The following five objective evaluation metrics are introduced in this article to measure the similarity between each detection result and the GT, including precision (Pre), recall ($R$), overall accuracy (OA), F1-score ($F1$), and Kappa.

TP is true positive, denoting the number of correctly predicted changed pixels; FP is false positive, denoting the number of unchanged pixels that are predicted as changed ones; FN is false negative, denoting the number of changed pixels that are predicted as unchanged ones; and TN is true negative, denoting the number of correctly predicted unchanged pixels.

The higher the Pre value, the lower the false alarm rate; the higher the $R$ value, the lower miss alarm rate. OA is the ratio of correctly predicted samples; $F1$ is the harmonic mean of Pre and $R$. Furthermore, Kappa is employed to address the insensitivity of OA due to class imbalance

$$\begin{cases} \text{Pre} = \text{TP}/(\text{TP}+\text{FP}) \\ R = \text{TP}/(\text{TP}+\text{FN}) \\ F1 = (2\times\text{Pre}\times R)/(\text{Pre}+R) \\ \text{Kappa} = (\text{OA}-P)/(1-P). \end{cases} \tag{9}$$

$P$, as an intermediate variable for calculating Kappa, is known as "random consistency error" or "chance coincidence rate." It is calculated as follows:

$$P = \frac{(\text{TP}+\text{TN})(\text{TP}+\text{FP})+(\text{FN}+\text{FP})(\text{FN}+\text{TN})}{(\text{TP}+\text{TN}+\text{FP}+\text{FN})^2}. \tag{10}$$

### B. Comparison Algorithms and Experimental Environment

*1) Comparison Algorithms:* In this article, five representative and open-source attention-based networks with Siamese structure framework, published in the past three years, are selected as comparison algorithms. The specific details of these algorithms are given as follows.

TABLE I
COMPARISON OF METRICS RESULTS FOR THE LEVIR DATASET

| Method | *Pre* | *R* | *OA* | *F1* | *Kappa* |
|---|---|---|---|---|---|
| STANet | 0.7652 | **0.9149** | 0.9814 | 0.8334 | 0.8236 |
| DR-TANet | 0.9112 | 0.8725 | 0.9892 | 0.8914 | 0.8857 |
| SNUNet | 0.8715 | 0.8883 | 0.9876 | 0.8798 | 0.8733 |
| SRCDNet | 0.6358 | 0.7665 | 0.9657 | 0.6950 | 0.6771 |
| DSAMNet | 0.8136 | 0.8927 | 0.9841 | 0.8513 | 0.8429 |
| GeSANet | **0.9204** (↑0.92%) | 0.8815 (↓3.34%) | **0.9901** (↑0.09%) | **0.9005** (↑0.91%) | **0.8953** (↑0.96%) |

*STANet (2020) [19]:* A network uses a multiscale pyramid attention mechanism to model the spatial–temporal relationship between any two pixels at different positions.

*SNUNet (2022) [24]:* The NestedUNet and ECAM are combined organically to create a framework that can transfer shallow fine-grained pixel-level information to the output, solving the issue of low target edge detection accuracy caused by continuous downsampling operations.

*DR-TANet (2021) [26]:* A dynamic receptive temporal attention module is proposed, which calculates the spatial–temporal joint similarity between bitemporal feature maps, aiming to explore the relationship between the size of the temporal dependent range and the detection performance. The horizontal and vertical attention improvement module is further combined to enhance the detection accuracy for bar objects.

*SRCDNet (2021) [39]:* To solve the problem of inconsistency in resolution between bitemporal images, the stacked attention module is used for capturing channel–spatial joint features in a multiscale feature map.

*DSAMNet (2022) [40]:* Each level feature map of the backbone is supplemented with the channel–spatial joint attention module, and the intermediate results are then employed to calculate the hybrid loss to further supervise the prediction results.

*2) Experimental Environment:* The experiment is implemented by the PyTorch1.2 framework under Python3.6. The training and testing are completed on an Intel Xeon[1] workstation with one NVIDIA Titan X (Pascal) GPU. The patch size of input is fixed to 256, using pretrained ResNet18 [41] as a backbone. The Adam optimizer [42] is used to train the network, and the learning rate is set to 0.001 with $\beta 1 = 0.9$ and $\beta 2 = 0.999$, which are used for computing running averages of gradient and its square. The 180 epochs are specified to end the training. Due to the limitation of memory size, the batch size of STANet can only be set to 4. In order to ensure the fairness of the experiments, the batch size is also fixed to 4 for all other experiments, and the remaining hyperparameters are consistent with those in the original paper.

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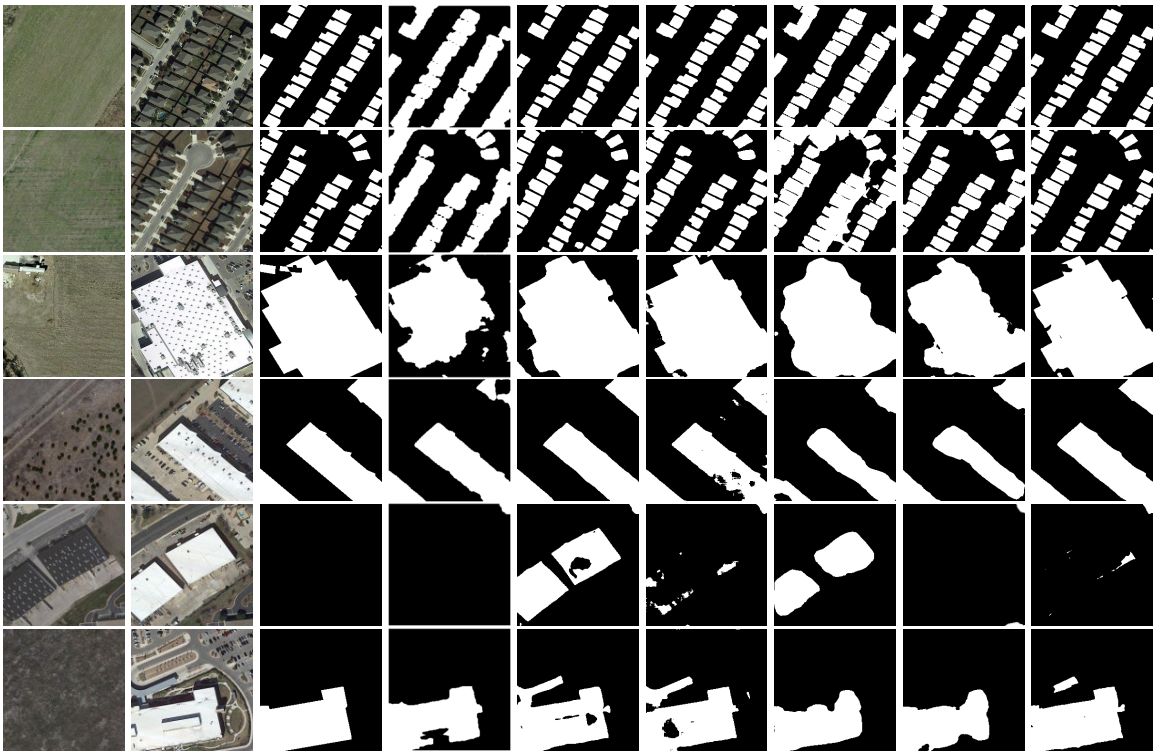



Fig. 6. Comparison results for LEVIR test samples with large area change detection. (a) T1. (b) T2. (c) GT. (d) STANet. (e) DR-TANet. (f) SNUNet. (g) SRCDNet. (h) DSAMNet. (i) GeSANet.

### C. LEVIR Experiment Results and Analysis

Table I shows the objective evaluation results of the LEVIR dataset. The SRCDNet obtains the lowest performance on all five metrics, and the gap is distinct compared with other methods, revealing that this network is almost difficult to converge when the batch size is small. Subsequently, both STANet and DSAMNet receive lower Pre and higher $R$, indicating that these two networks are not strong enough in overcoming pseudo-change information. The other two comparison methods and GeSANet perform well on all five metrics. In particular, DR-TANet, which has won the second place, demonstrates that the exploration of temporal correlation between multiscale feature maps is necessary. Relying on simple subtraction or concatenation operation for measurement is insufficient. The proposed GeSANet achieves the first-place method Pre, OA, $F1$, and Kappa indices, which are 0.92%, 0.09%, 0.91%, and 0.96% higher than the second-place method, respectively.

The subjective comparisons for LEVIR are given in Figs. 6 and 7, where Fig. 6 shows the CD with large areas and Fig. 7 shows the CD with small targets. As can be seen from Fig. 6, the proposed GeSANet can detect not only the location, shape, and size information of the change entities more accurately but also more detailed at the edge portion. The GeSANet has excellent capability in recognizing pseudo-change information caused by various factors.

From Fig. 7, it can be observed that SRCDNet and DSAMNet, in this case, do not achieve ideal results, while STANet performs well and can detect the position of change entities accurately. The detection results of DR-TANet and SNUNet are still convincible. Also, the GeSANet performs well in detecting small change targets as well.

To observe the difference in detecting edge details more intuitively, the pseudo-color visualizations of misdetected pixels are shown in Fig. 8. The pixels colored green represents false alarms, and the pixels colored red represents miss alarms.

### D. CDD Experiment Results and Analysis

Table II presents the objective evaluation results for the CDD dataset. The proposed method has achieved optimal results except for the $R$ index. The $F1$ and Kappa indices are 4.83% and 5.55% higher than the second-place DSAMNet, and the $R$ index is only 0.55% different from the first place. The performance of SRCDNet is still not satisfactory; STANet remains the highest $R$ value and a bit lower Pre value, and on the contrary, SNUNet achieves the second highest Pre value and a bit lower $R$ value, which means that the former has a high false alarm rate and the latter has a high miss alarm rate. The DR-TANet and DSAMNet play a stable detection performance.

Figs. 9 and 10 show the subjective comparison results of CDD test samples. Similarly, we finely divide the comparisons into two groups, one for large area change scenes (Fig. 9) and the other for small target change scenes (Fig. 10).

From Fig. 9, it can be clearly observed that the detection results are consistent with Table II. The backgrounds of SNUNet detection results are relatively messy, proving that when change scenes are extremely complex and diverse, the introduction of too many shallow pixel-level features will lead the network unstable. Inversely, SRCDNet has many miss

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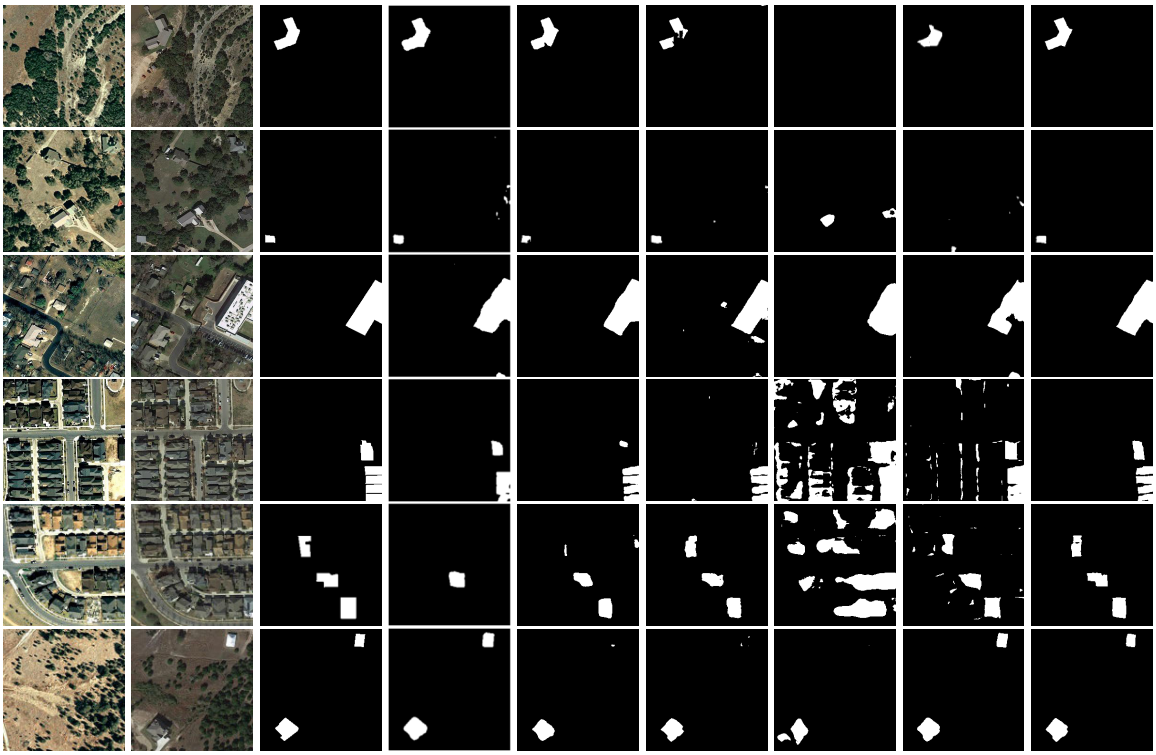



Fig. 7. Comparison results for LEVIR test samples with small target change detection. (a) T1. (b) T2. (c) GT. (d) STANet. (e) DR-TANet. (f) SNUNet. (g) SRCDNet. (h) DSAMNet. (i) GeSANet.

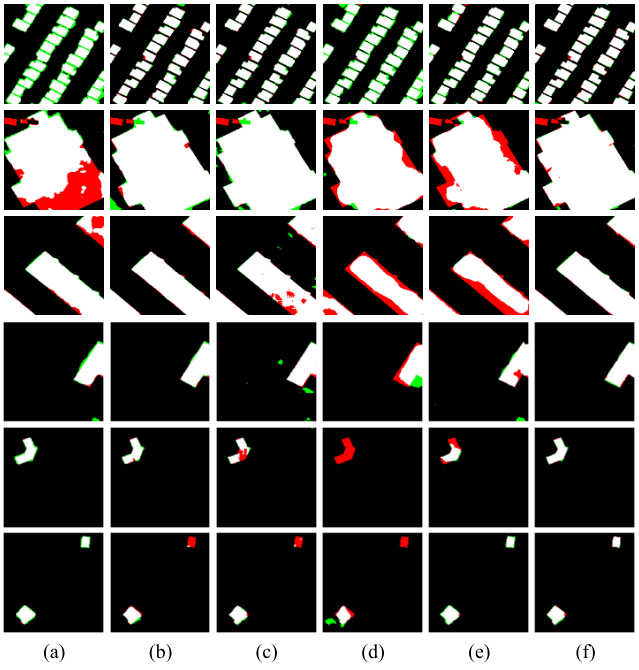



Fig. 8. False alarms (colored green) and miss alarms (colored red) for LEVIR test samples. (a) STANet. (b) DR-TANet. (c) SNUNet. (d) SRCDNet. (e) DSAMNet. (f) GeSANet.

alarms; STANet and DR-TANet have some shortcomings in edge detections. For example, STANet has a large number of false alarms in the 3rd and 4th samples, and DR-TANet is discontinuous at the edge of Samples 1 and 2. GeSANet has more accurate detection results and more precise detection edges, which presents that it can reliably discriminate real change information and filter out various pseudo-change information.

TABLE II
COMPARISON OF METRICS RESULTS FOR THE CDD DATASET

| Method | *Pre* | *R* | *OA* | *F1* | *Kappa* |
|---|---|---|---|---|---|
| STANet | 0.8303 | **0.9464** | 0.9708 | 0.8845 | 0.8679 |
| DR-TANet | 0.9352 | 0.8615 | 0.9755 | 0.8968 | 0.8830 |
| SNUNet | 0.9467 | 0.8054 | 0.9704 | 0.8703 | 0.8537 |
| SRCDNet | 0.7608 | 0.6659 | 0.9329 | 0.7102 | 0.6725 |
| DSAMNet | 0.8923 | 0.9141 | 0.9758 | 0.9031 | 0.8892 |
| GeSANet | **0.9623** (↑1.56%) | 0.9409 (↓0.55%) | **0.9883** (↑1.25%) | **0.9514** (↑4.83%) | **0.9447** (↑5.55%) |

Fig. 10 supplies the change detection results for small target change samples in the CDD dataset, and the results show that the detection accuracy of SNUNet and SRCDNet for small targets is poor, presenting as serious miss alarms, especially the changes with long edges. At the same time, the performances of DR-TANet in the detection of some samples are not ideal, either. On the contrary, STANet is able to detect the shape of change entities, which is consistent with the data in Table II. Nevertheless, the capability to locate change edges is insufficient as well. For example, it results in a high false alarm rate in Samples 1 and 3; DSAMNet and GeSANet gain more robust detection performance, but DSAMNet has the insufficient capability to suppress pseudo-change information, which makes it believe that there is a changed entity on the left side of Sample 1.

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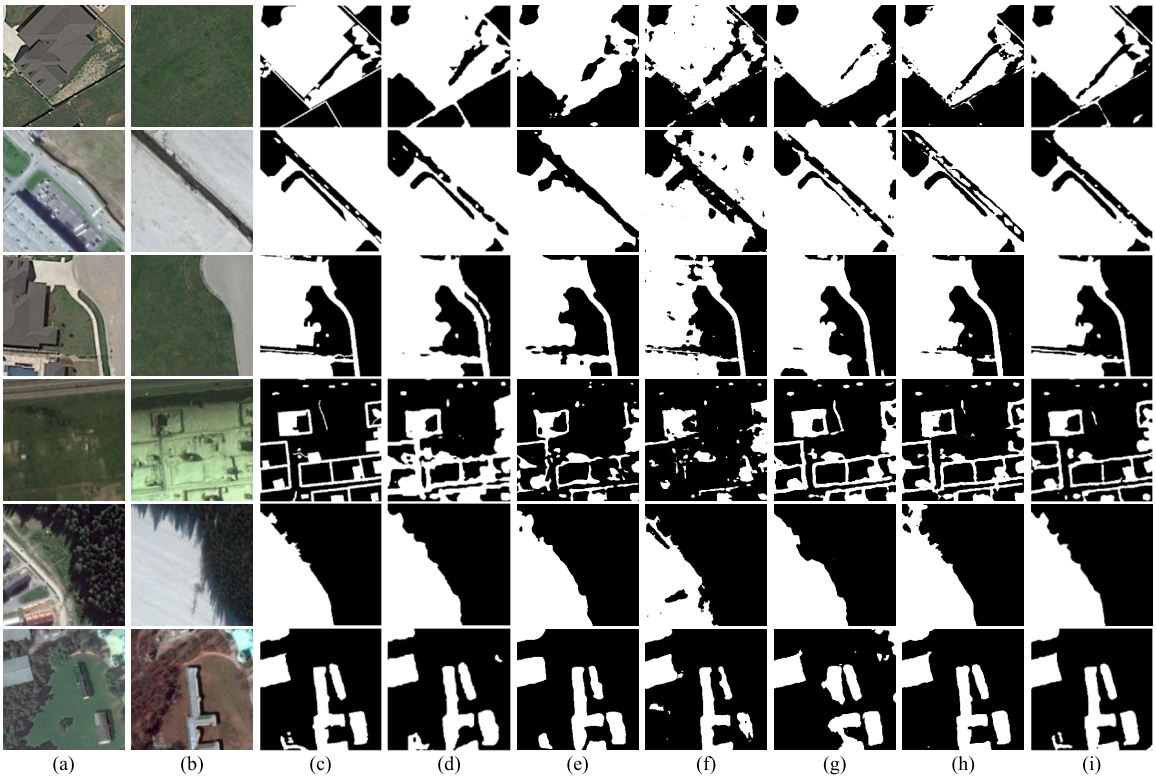



Fig. 9. Comparison results for CDD test samples with large area change detection. (a) T1. (b) T2. (c) GT. (d) STANet. (e) DR-TANet. (f) SNUNet. (g) SRCDNet. (h) DSAMNet. (i) GeSANet.

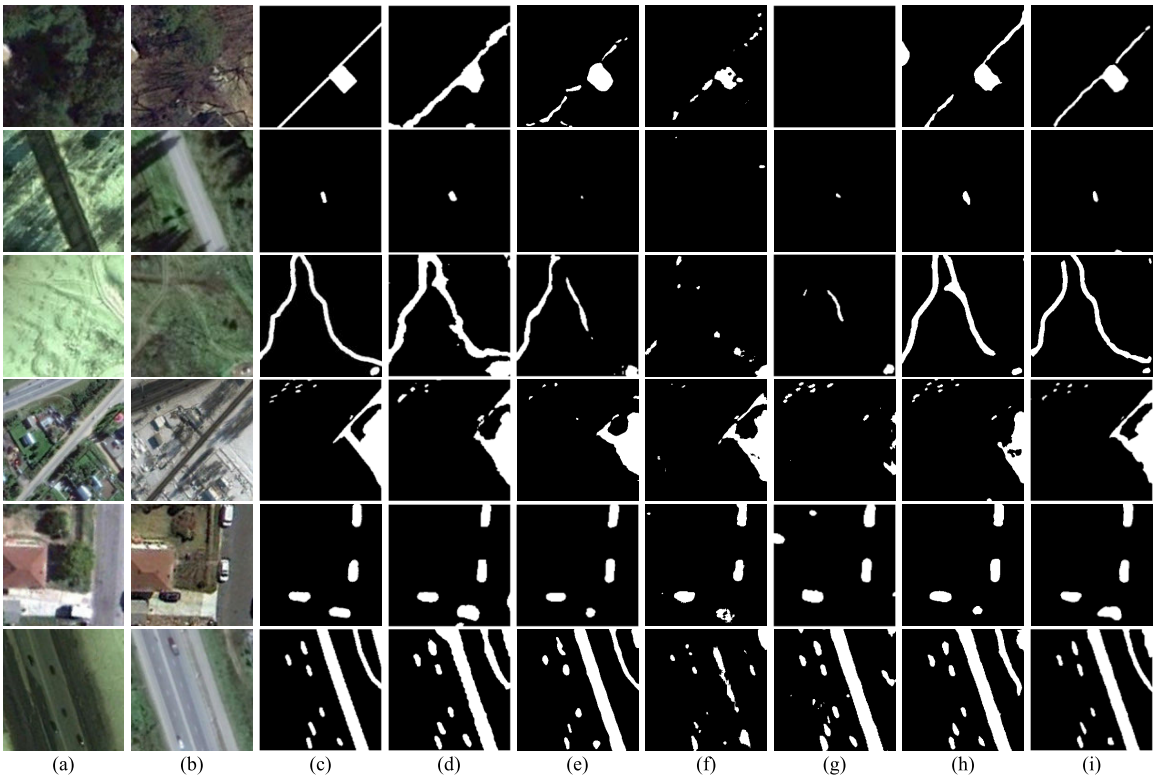



Fig. 10. Comparison results for CDD test samples with small target change detection. (a) T1. (b) T2. (c) GT. (d) STANet. (e) DR-TANet. (f) SNUNet. (g) SRCDNet. (h) DSAMNet. (i) GeSANet.

Fig. 11 shows the misdetected pixels in Figs. 9 and 10. It can be seen that except GeSANet, all the other methods have a large number of false alarms (colored green) and miss alarms (colored red) in each test sample, especially for the detection

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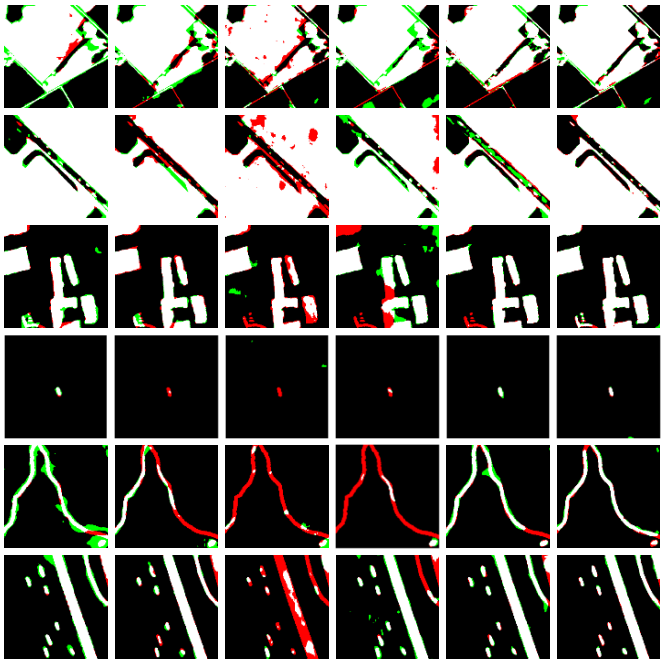



Fig. 11. False alarms (colored green) and miss alarms (colored red) for CDD test samples. (a) STANet. (b) DR-TANet. (c) SNUNet. (d) SRCDNet. (e) DSAMNet. (f) GeSANet.

TABLE III
OBJECTIVE METRIC COMPARISONS OF ABLATION EXPERIMENTS

| LEVIR | | | | | |
|---|---|---|---|---|---|
| **Index** | **Baseline** | **PMM** | **CRM** | **w/o Res** | **GeSANet** |
| ***Pre*** | 0.9175 | 0.9157 (↓**0.18%**) | 0.9128 (↓0.47%) | 0.9154 | **0.9204** |
| ***R*** | 0.8753 | 0.8821 (↑0.58%) | 0.8828 (↑**0.75%**) | **0.8853** | 0.8815 |
| ***OA*** | 0.9896 | 0.9899 (↑**0.03%**) | 0.9897 (↑0.01%) | 0.9900 | **0.9901** |
| ***F1*** | 0.8959 | 0.8986 (↑**0.27%**) | 0.8976 (↑0.17%) | 0.9001 | **0.9005** |
| ***Kappa*** | 0.8905 | 0.8932 (↑**0.27%**) | 0.8922 (↑0.17%) | 0.8949 | **0.8953** |
| CDD | | | | | |
| **Index** | **Baseline** | **PMM** | **CRM** | **w/o Res** | **GeSANet** |
| ***Pre*** | 0.9344 | 0.9187 (↓1.57%) | 0.9505 (↑**1.61%**) | 0.9539 | 0.9623 |
| ***R*** | 0.8128 | 0.9094 (↑9.66%) | 0.9285 (↑**11.57%**) | 0.9266 | 0.9409 |
| ***OA*** | 0.9698 | 0.9789 (↑0.91%) | 0.9852 (↑**1.54%**) | 0.9854 | 0.9882 |
| ***F1*** | 0.8693 | 0.9140 (↑4.47%) | 0.9394 (↑**7.01%**) | 0.9400 | 0.9514 |
| ***Kappa*** | 0.8523 | 0.9020 (↑4.97%) | 0.9310 (↑**7.87%**) | 0.9317 | 0.9447 |

at edges. The performance of GeSANet is far superior to other methods on the CDD dataset.

### *E. Ablation Experiments*

To verify the effect of each part in GeSANet, the corresponding ablation experiments are carried out in this section. The following four experiments are included.

1) *Baseline:* The experiment uses the difference to replace PMM and remove CRM part.
2) *PMM:* The experiment only with PMM.
3) *CRM:* The experiment only with CRM.
4) *w/o Res:* The experiment with both PMM and CRM, but connected without $F_{\text{PMM}}$ residual learning.
5) *GeSANet:* The proposed GeSANet.

Table III shows the comparison of objective metric results of each ablation experiment for both LEVIR and CDD datasets. Generally speaking, except the Pre indexes decrease sometimes, the rest have increased to a certain extent, indicating that each part has a certain contribution to GeSANet, especially in the improvement of the $R$, $F1$, and Kappa indexes, which greatly reduces the false alarm rate.

From the enhancement numerical values of baseline to each model given in Table III, for the LEVIR dataset, the CRM has a smaller improvement on the baseline, but the PMM has a larger improvement on baseline (bold part), verifying that PMM plays a more important role in this experiment, that is, PMM can correct misregistration in geospatial position. For the CDD dataset, the PMM has a smaller improvement on the baseline, but the CRM has a larger improvement on baseline (bold part), indicating that CRM plays a more important role in this experiment, verifying that in extremely complex detection background, various pseudo-change information can be effectively filtered out through reasoning and effective real change information can be reconstructed at the same time.

Fig. 12 shows the visual comparisons of ablation results on LEVIR test samples and false alarms (colored green) and miss alarms (colored red) in each detection result. It can be observed from Sample 1 that GeSANet has fewer green pixels compared with other ablation experiments, which suggests that GeSANet effectively restrains the interference of pseudo-change information. Samples 2 and 3 demonstrate the low false alarm rate of the proposed method, that is, there are fewer red pixels. Sample 4 demonstrates that it is significantly more precise than the other experiments in detecting edges. Samples 5 and 6 illustrate that PMM and CRM sometimes complement with each other in small target detections.

Fig. 13 presents the visual comparison of ablation results on CDD test samples. The results reveal that compared with other ablation experiments, the different results of GeSANet have fewer red and green pixels, representing that GeSANet combines the detection advantages of PMM and CRM, with both lower false alarm rate and miss alarm rate. It can not only overcome pseudo-change information caused by multiple factors such as season alternating and shadow changing but also effectively detect the changing entities with line shape.

Finally, Table IV shows the comparison of parameters and flops for each method with one batch size and also provides the test time for one test sample. The experiment is set up on a workstation with Intel Xeon E5-2630 CPU and one NVIDIA Titan X (Pascal) GPU. Each test sample is with 256 × 256 patch size. It can be seen that SRCDNet and DSAMNet are with smaller numbers of parameters, but with large flops, while DR-TANet and the proposed GeSANet are with larger numbers of parameters, but with small flops. After statistical calculations, it illustrates that the PAM in

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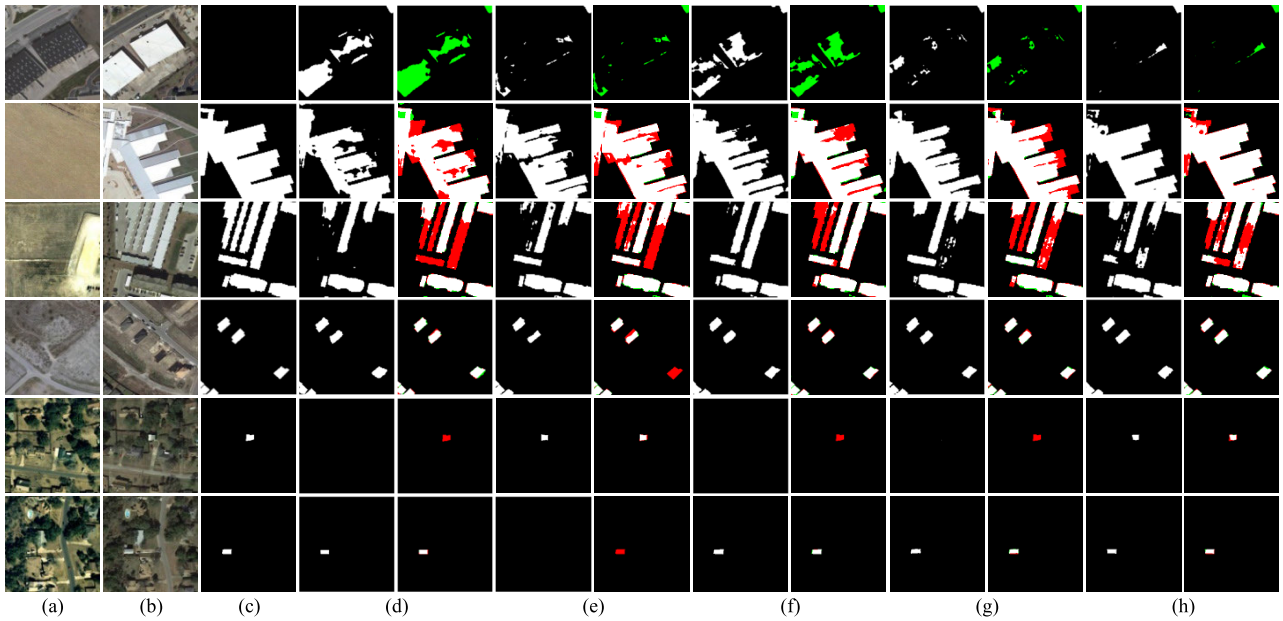



Fig. 12. Visual comparison of ablation results on LEVIR test samples. (a) T1. (b) T2. (c) GT. (d) Baseline. (e) PMM. (f) CRM. (g) w/o Res. (h) GeSANet.

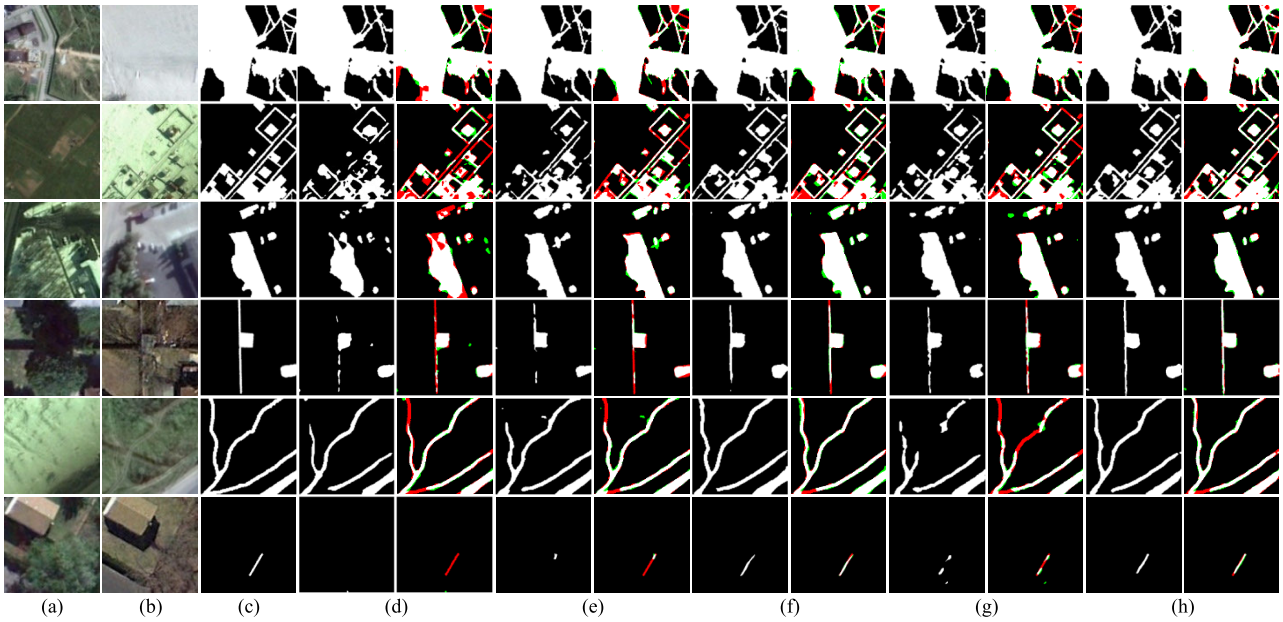



Fig. 13. Visual comparison of ablation results on CDD test samples. (a) T1. (b) T2. (c) GT. (d) Baseline. (e) PMM. (f) CRM. (g) w/o Res. (h) GeSANet.

STANet has 1.216 Gflops and the DRTAM in DR-TANet has 2.287 Gflops, while the number of flops of our CRM is only 0.078 G, and the number of parameters is only 0.791 M, which is less than the 1.045 M of DRTAM. The previous experimental results show that the proposed method can obtain better detection results and higher timeliness within the same training time. Also, from the perspective of test time, the efficiency of the proposed network is also satisfactory.

TABLE IV
COMPARISON OF PARAM. AND FLOPS FOR EACH ALGORITHM

| Method | STA-Net | DR-TANet | SNU-Net | SRCD-Net | DSAM-Net | GeSA-Net |
|---|---|---|---|---|---|---|
| Param. (M) | 16.929 | 33.390 | 27.068 | 16.980 | 16.951 | 36.503 |
| Flops (G) | 6.734 | 7.073 | 123.160 | 71.675 | 75.388 | 7.257 |
| Test time (s) | 0.0206 | 0.0199 | 0.0232 | 0.0238 | 0.0208 | 0.0203 |

### *F. Discussion*

*1) Effects Analysis of the PMM on Registration:* The proposed PMM aims to reduce the error accumulation caused by the misregistration error between pre- and post-temporal images on change detection results. One could suspect whether the improvement is due to the registration effect of PMM or just introducing additional learnable parameters.

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TABLE V
OBJECTIVE METRIC COMPARISONS OF PMM EXPERIMENTS

| LEVIR | | | | | |
|---|---|---|---|---|---|
| **Index** | ***Pre*** | ***R*** | ***OA*** | ***F1*** | ***Kappa*** |
| PMM | 0.9204 | 0.8815 | 0.9901 | 0.9005 | 0.8953 |
| Modified PMM | 0.9179 | 0.8806 | 0.9899 | 0.8989 | 0.8937 |

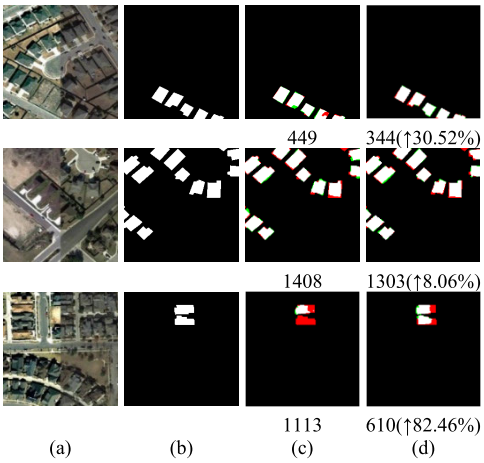



Fig. 14. False alarms (colored green) and miss alarms (colored red) for LEVIR misregistration test samples. (a) T1 (left) and T2 (right). (b) GT. (c) Modified PMM. (d) PMM.

To validate the above concern, a special experiment is conducted. The order of subtraction (refer to Fig. 3) in PMM is brought forward (named "modified PMM" in Table V), that is, calculating the difference between bitemporal feature maps first and then performing the same feature extraction manner, making the deformable convolution conducted on the change features. Since the misregistration problem is more prominent in the LEVIR dataset, Table V reports the objective metric comparisons of corresponding experiments for the LEVIR dataset. It can be seen that all indicators have declined.

In this way, the registration effect of PMM can be validated without increasing the number of learnable parameters. Fig. 14 shows the visual comparisons of the corresponding experiments. The effectiveness of PMM in reducing geospatial differences between bitemporal images can be observed by counting the number of error-detected pixels in each change map.

## V. CONCLUSION

In this article, the geospatial PMM and geospatial CRM are designed to solve the problems of heavy dependence on registration and insufficient suppression of diverse pseudo-change information, which occurs frequently in VHR RSIs, and the GeSANet is further proposed. For PMM, it breaks the tradition of the original convolutional operation upon a regular grid by assigning independent offsets to each position in the feature map, so as to make use of subpixel information to correct spatial misregistration. For CRM, tensor completion theory is introduced to reason real change information from the corrected sparse feature map, to achieve the purpose of filtering out pseudo-change information and further enhancing the discrimination capability of the network. Numerous experimental visualization results prove that the proposed GeSANet is effective and robust.

## REFERENCES

[1] R. Selvaraj and S. Nagarajan, "Change detection techniques for a remote sensing application: An overview," in *Cognitive Systems and Signal Processing in Image Processing*. U.K.: Elsevier, 2022, pp. 129–143.
[2] S. Han and J. P. Kerekes, "Overview of passive optical multispectral and hyperspectral image simulation techniques," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 11, pp. 4794–4804, Nov. 2017.
[3] Y. You, J. Cao, and W. Zhou, "A survey of change detection methods based on remote sensing images for multi-source and multi-objective scenarios," *Remote Sens.*, vol. 12, no. 15, pp. 2460–2499, 2020.
[4] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.
[5] W. Zhang, X. Lu, and X. Li, "A coarse-to-fine semi-supervised change detection for multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3587–3599, Jun. 2018.
[6] J. Chen, X. Chen, X. Cui, and J. Chen, "Change vector analysis in posterior probability space: A new method for land cover change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 2, pp. 317–321, Mar. 2011.
[7] L. I. Kuncheva and W. J. Faithfull, "PCA feature extraction for change detection in multidimensional unlabeled data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 1, pp. 69–80, Jan. 2014.
[8] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.
[9] C. Wu, L. Zhang, and B. Du, "Kernel slow feature analysis for scene change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2367–2384, Apr. 2017.
[10] T. Habib, J. Inglada, G. Mercier, and J. Chanussot, "Support vector reduction in SVM algorithm for abrupt change detection in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 3, pp. 606–610, Jul. 2009.
[11] J. Gong, X. Hu, S. Pang, and K. Li, "Patch matching and dense CRF-based co-refinement for building change detection from bi-temporal aerial images," *Sensors*, vol. 19, no. 7, pp. 1557–1585, 2019.
[12] M. L. F. Velloso, M. Simoes, and T. A. Carneiro, "Competitive artificial neural network for change-detection of land cover: An unsupervised approach," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Toronto, ON, Canada, Sep. 2002, pp. 95–97.
[13] V. E. Neagoe, N. Mihai, and D. Mihai, "Neural network techniques for automated land-cover change detection in multispectral satellite time series imagery," *Int. J. Math. Models Methods Appl. Sci.*, vol. 6, no. 1, pp. 130–139, 2012.
[14] X. Huang et al., "A multispectral and multiangle 3-D convolutional neural network for the classification of ZY-3 satellite images over urban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10266–10285, Dec. 2021.
[15] K. Kamirul, W. Hasbi, P. R. Hakim, and A. H. Syafrudin, "Automatic ship recognition chain on satellite multispectral imagery," *IEEE Access*, vol. 8, pp. 221918–221931, 2020.
[16] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
[17] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
[18] A. Luciano et al., *Remote Sensing Image Fusion*. Boca Raton, FL, USA: CRC Press, 2015.
[19] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, pp. 1662–1684, 2020.
[20] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 565–571, May 2018.
[21] C. Wu, L. Zhang, and L. Zhang, "A scene change detection framework for multi-temporal very high resolution remote sensing images," *Signal Process.*, vol. 124, pp. 184–197, Jul. 2016.

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[22] X. Zhang et al., "Multimodal remote sensing image registration methods and advancements: A survey," *Remote Sens.*, vol. 13, no. 24, pp. 5128–5158, 2021.
[23] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
[24] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[25] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.
[26] S. Chen, K. Yang, and R. Stiefelhagen, "DR-TANet: Dynamic receptive temporal attention network for street scene change detection," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jul. 2021, pp. 502–509.
[27] Q. Ke and P. Zhang, "CS-HSNet: A cross-Siamese change detection network based on hierarchical-split attention," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9987–10002, 2021.
[28] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.
[29] N. Shi, K. Chen, and G. Zhou, "A divided spatial and temporal context network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4897–4908, 2022.
[30] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.
[31] K. Song, F. Cui, and J. Jiang, "An efficient lightweight neural network for remote sensing image change detection," *Remote Sens.*, vol. 13, no. 24, p. 5152, Dec. 2021.
[32] X. Wang, K. C. K. Chan, K. Yu, C. Dong, and C. C. Loy, "EDVR: Video restoration with enhanced deformable convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Long Beach, CA, USA, Jun. 2019, pp. 1954–1963.
[33] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," 2022, *arXiv:2209.08290*.
[34] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 764–773.
[35] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 9300–9308.
[36] Q. Zhao, L. Zhang, and A. Cichocki, "Bayesian CP factorization of incomplete tensors with automatic rank determination," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1751–1763, Sep. 2015.
[37] M. A. Veganzones, J. E. Cohen, R. C. Farias, J. Chanussot, and P. Comon, "Nonnegative tensor CP decomposition of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2577–2588, May 2016.
[38] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.
[39] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403718.
[40] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
[41] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
[42] D. P. Kingma and J. Ba, "ADAM: A method for stochastic optimization," in *Proc. ICLR*, San Diego, CA, USA, May 2015, pp. 1–12.



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